

N100 Financial Intelligence Platform

Analyst Guide

A financial analytics platform for screening, comparing and analysing NIFTY 100 companies.

This guide explains the dashboard, API, screening workflow, company analysis, peer comparison, valuation analysis and troubleshooting.

1. Platform Overview

The platform combines financial data, computed KPIs, screening rules, peer analysis, portfolio statistics and company-level analytics.

The system contains data for 92 companies and exposes the analytics through a Streamlit dashboard and FastAPI service.

The primary users are financial analysts, researchers and investors who need structured financial intelligence.

2. Starting the Application

Activate the Python virtual environment before running the project.

Start FastAPI using: `uvicorn src.api.main:app --reload --port 8000`

Start Streamlit using the project's dashboard application and port 8501.

FastAPI provides the backend data services while Streamlit provides the analyst-facing interface.

3. Company Profile

Open the Company Profile screen and select a company ticker.

The profile provides company information and latest financial KPIs.

Historical P&L, balance sheet, cash flow and ratio information can be inspected through the company APIs.

TCS is available as a validated example company.

4. Financial Statements

The P&L; screen provides historical sales, expenses, operating profit, margins, interest, depreciation, profit before tax, tax and net profit.

The balance sheet provides equity, reserves, borrowings, liabilities, fixed assets, investments and total assets.

The cash flow view provides operating, investing, financing and net cash flow information.

Historical endpoints support year-based filtering.

5. Financial Ratios and KPIs

The platform computes financial KPIs including profitability, leverage, valuation and cash-flow metrics.

ROE measures return generated relative to shareholder equity.

Debt-to-equity provides a leverage perspective.

Interest coverage indicates the ability to service interest obligations.

Cash-flow KPIs provide additional information about earnings quality and capital allocation.

6. Screener

Use the Screener to filter companies according to financial criteria.

Available filters include minimum ROE, maximum debt-to-equity, minimum free cash flow, sector, revenue CAGR, PAT CAGR and maximum P/E.

Results are ranked and returned with the relevant company metrics.

Invalid parameter values are rejected by the API.

7. Sector Analysis

The sector endpoint provides sector-level company counts and financial statistics.

The platform currently contains 10 populated broad sectors in the database.

Sector company views allow analysts to inspect companies belonging to a selected sector.

Information Technology is one of the available sectors.

8. Peer Comparison

Peer analysis groups comparable companies for relative evaluation.

Peer endpoints provide companies belonging to a peer group and percentile information across financial metrics.

The peer comparison view provides radar-style comparative data for a selected company, peer average and benchmark.

9. Market Capitalisation and Valuation

Market-cap data contains historical market capitalisation and enterprise value.

Valuation metrics include P/E, P/B, EV/EBITDA and dividend yield.

The market-cap endpoint exposes historical valuation information for supported companies.

10. Portfolio Statistics

Portfolio statistics provide percentile distributions across the company universe.

The API returns P10 through P90 values for 10 core KPIs.

These statistics help analysts understand where an individual company sits relative to the broader universe.

11. API Usage

The API is available under the /api/v1 namespace.

Examples:

GET /api/v1/health

GET /api/v1/companies

GET /api/v1/companies/TCS

GET /api/v1/companies/TCS/pl

GET /api/v1/companies/TCS/bs

GET /api/v1/companies/TCS/cashflow

GET /api/v1/companies/TCS/ratios

GET /api/v1/screener

12. Tearsheet Generation

Company tearsheets are provided as PDF documents.

The tearsheet API returns application/pdf content.

A typical request is GET /api/v1/companies/TCS/tearsheet.

The generated document can be saved or opened using a standard PDF viewer.

13. Troubleshooting

If the API cannot be reached, verify that FastAPI is running on port 8000.

If the dashboard cannot load data, verify that the API is running and the database exists at db/nifty100.db.

If an endpoint returns 404, verify the ticker or sector name.

If tests fail during collection, verify the virtual environment and required Python packages.

For API tests using FastAPI TestClient, ensure the required HTTP client dependency is installed.

14. Testing and Quality Assurance

The project contains ETL, KPI, DQ and API tests.

The completed test suite currently passes 180 tests.

The API performance test completed 10 concurrent screener requests in approximately 0.14 seconds.

Five Company Profile API workflows were tested successfully below the three-second requirement.

15. Analyst Workflow

Start with the Screener to identify candidate companies.

Open Company Profile for detailed financial analysis.

Review historical financial statements and ratios.

Compare the company with relevant peers.

Review valuation and market-cap information.

Use portfolio statistics for universe-level context.

Generate a tearsheet when a standalone company report is required.